

“... conical ball... pass through the bodies of two men and lodge in the body of a third...”

ATTRIBUTED TO **GEORGE MACLEOD, CRIMEAN WAR SURGEON**

enemy, now danger lay up to a distance of 1,000 yards (914m) or more. In the US, the new Model 1855 Springfield rifle employed the Minié bullet, while in Britain, the first rifle to use the new bullet on a large scale was the Enfield Pattern 1853 (see pp.100–01). In the Crimean War, it was discovered that with these rifles, for the first time, infantry could outgun artillery, picking off the gunners from a safe distance. A few years later, almost a million

Pattern 1853 rifles would be shipped to serve both sides in the American Civil War. Battles, once close-quarter volleys followed by tides of bayonet or cavalry charges, now became long-range engagements from entrenched positions, against which a cavalry charge was almost suicidal. Judgment of distance and setting of sights now became paramount in making the rifle, in the hands of well-trained infantry, the new god of the battlefield.

KEY FIGURE

CLAUDE-ETIENNE MINIÉ
(1804–79)

Claude-Etienne Minié served as captain with the French Chasseurs (light infantry) in North Africa. He was frustrated with the shortcomings of the muskets issued to his troops. Following his invention of the Minié bullet, he was awarded 20,000 French francs and made an instructor at the Vincennes military establishment. In 1858, he retired as colonel, later becoming a military instructor for the Khedive of Egypt, and then manager at the Remington Arms Company, US.



AFTER >>

The Minié bullet was critical in spurring on the development of long-range shooting. New military training regimes were needed. National Rifle Associations, such as those formed in Britain and the US, encouraged long-range target shooting as sport. Military sharpshooters became snipers—unseen long-range killers adding new levels of terror to an already fearsome business.

• **MILITARY TACTICS** had to be revised in the face of long-range accuracy, since close-range combat would increase the likelihood of soldiers being killed.

• **INDIVIDUAL SHARPSHOOTERS** and snipers picking off specific targets replaced the military tradition of “firing by numbers,” or volley-fire.

• **DEADLY TEAMS OF SNIPERS** and “spotters” evolved; the spotters used telescopes to identify targets and passed details to the snipers.

• **HIGHER-VELOCITY BULLETS** inflicted greater damage than earlier bullets. Instead of repairable wounds to arms and legs, amputations became common.

• **NEW SNIPER RIFLES** in the 20th century, firing a .50in machine-gun cartridge, made it possible to aim at and hit human targets at ranges of more than 1 mile (1.7km), far greater than the ½-mile (0.9-km) range of an early muzzle-loading rifle.



**.50IN BMG
CARTRIDGE,
1910**

